

Diagnosing Identifiability in Two-Tower Models for Unbiased Learning to Rank

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Abstract

Two-tower models are a popular unbiased learning-to-rank (ULTR) approach for mitigating position bias in clicks. A major challenge in two-tower models is identifiability: whether relevance and position bias can be uniquely determined from clicks. Recent work shows that two-tower models can be identified when the same documents (or documents with similar features) are observed across positions. However, these conditions are defined in infinite data. In practice, we lack methods for diagnosing identifiability in real-world datasets with small sample sizes and limited positional variability.

In this work, we present a practical identifiability diagnostic for two-tower models. We quantify whether individual bias parameters are uniquely determined by shifting them away from their optimal values, retraining parts of the model, and measuring whether click-prediction performance degrades significantly. Identified parameters exhibit measurable performance loss when shifted, while unidentified parameters can be freely adjusted without affecting model fit. Our method can distinguish between cases in which identifiability fails due to limitations in model assumptions or data collection and cases caused by a limited sample size. We demonstrate, through simulation, how non-deterministic logging policies, feature overlap, and increased sample size affect identifiability. Applying our method to the real-world Baidu-ULTR dataset, we find that despite large amounts of click data, some parameters of two-tower models remain unidentified, highlighting the practical need for diagnostic methods. All code, data, and results are available at: <https://github.com/stanfris/practical-identifiability-ultr>

CCS Concepts

• Information systems → Learning to rank.

Keywords

Unbiased learning to rank, Click models, Two-tower models

ACM Reference Format:

Stan Fris and Philipp Hager. 2026. Diagnosing Identifiability in Two-Tower Models for Unbiased Learning to Rank. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, July 20–24, 2026, Melbourne, VIC, Australia. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3805712.3809640>

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SIGIR '26, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-2599-9/2026/07

<https://doi.org/10.1145/3805712.3809640>

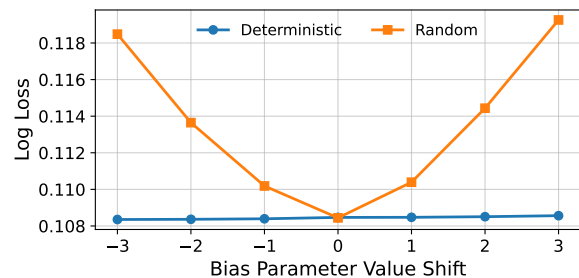


Figure 1: Profile likelihood for a single position bias parameter under deterministic and random logging policies. A flat curve indicates the parameter is unidentified, while a clear optimum indicates identification.

1 Introduction

Unbiased Learning to Rank (ULTR) uses implicit feedback, such as clicks, to optimize ranking models for web search and recommendation [12, 16, 19, 34]. Clicks, however, are known to be a biased signal of user preference [1, 17, 18, 28, 39, 42] and can be influenced by factors such as a document's position in the ranking [8, 17], its surrounding documents [3, 9, 42], or the user's trust in the search engine [1, 18, 32]. Therefore, clicks should not directly be used as positive or negative indicators of user preference [17].

In recent years, additive two-tower models have emerged as a method for addressing biases in click data, especially in industry settings [11, 15, 40, 41]. Two-tower models consist of two neural networks (or *towers*): a relevance tower that considers features indicating a document's relevance to a given search query, e.g., query and document text or traditional ranking features, such as BM25. The second network, or bias tower, estimates user examination based on contextual features that are not indicative of relevance but might affect clicks, such as position, device type (e.g., mobile or desktop), or the relative size of the document on screen [6, 40]. During training, the output of both towers is combined to predict clicks. But during inference, only the relevance tower is used to rank new items [11, 12, 14].

While these models have shown strong empirical performance [11, 12, 14], two-tower models rely on the key assumption that clicks can be factorized into a bias and a relevance signal. This decomposition faces a practical problem: if documents always appear in the same positions for a query, we cannot distinguish whether, e.g., a low click rate stems from low document relevance or high position bias [5, 27]. In this scenario, decreasing all relevance parameters while increasing bias parameters by the same amount yields identical click predictions, meaning that infinitely many model parameter combinations can explain the observed clicks equally well while

leading to very different conclusions about bias and relevance in a dataset. This inability to recover a unique set of parameters from observed clicks is the fundamental challenge of (un)identifiability in two-tower models [5, 14]. This challenge is typically resolved by observing the same document at multiple positions (a swap) for the same query [2, 8, 10] or by observing documents with similar features across positions (i.e., feature distributions overlap) [14, 35].

Prior work explores the conditions for identifying two-tower models theoretically in infinite data for very simple model architectures [5, 14]. Chen et al. [5] explore identifiability for models that learn a separate relevance parameter for each query-document pair. Hager et al. [14] extend this exploration to feature-based models and discuss that model configuration (e.g., network depth, or regularization) can affect identifiability. In practice, however, we lack ways to diagnose identifiability for a specific two-tower model architecture in real-world data, where issues such as small sample sizes, limited positional variability, feature distributions, and model capacity can affect identifiability [14]. For example, even if our production system randomly swaps documents across positions to guarantee model identifiability, our collected dataset might be too small to actually identify a model in practice. Therefore, in this work, we investigate how to diagnose identifiability and highlight the challenges posed by real-world datasets.

Specifically, we propose a method based on profile likelihood analysis, adopted from identifiability literature in bioinformatics [29]. Intuitively, the profile likelihood is a statistical method for investigating how sensitive a model’s fit is to changes in a single parameter [29, 36]. We can measure profile likelihood by shifting a single model parameter and observing the change in the entire model’s log-likelihood. Profile likelihood can also be used to test whether individual model parameters are uniquely determined: First, we train a model on a specific dataset and record its loss. Second, we shift a single model parameter to a different value before re-optimizing all remaining model parameters and recording the loss of the *shifted* model. If a model is identified, we should have a single optimal model parameter. Thus, when shifting a single parameter away from its optimum, we should observe that our model fits our observed data less well (i.e., we observe a higher loss). Using a confidence interval, we can determine whether the loss of the shifted model is significantly different from the loss of our original model. If, however, we observe an insignificant difference between the shifted model’s loss and our original loss, we have discovered two equally valid model parameterizations, meaning our model is not uniquely identified.

Additionally, profile likelihood also allows us to distinguish between different types of unidentifiability: *Structural unidentifiability* is a property of the model itself, arising from redundant parametrization where multiple parameter configurations produce identical predictions [29], which is also the lens used by previous work on two-tower models [5, 13, 27]. A structurally identified model, however, might still not be *practically identified* on a given dataset due to insufficient training data that is required to lead to a significant performance difference between different parameterizations [29, 37].

Figure 1 demonstrates profile likelihood as an identifiability diagnostic. We train two versions of a two-tower model on a synthetic click dataset with one-hot encoded features: one trained on random rankings, where documents are swapped freely across positions,

and one on deterministic rankings, where the logging policy always shows the same documents in the same positions. For the randomly-ranked model, shifting the first position bias parameter away from its optimum produces a clear increase in loss, indicating a unique solution. Under the deterministic policy, the same shift leaves the loss unchanged, revealing multiple equally valid parameterizations and confirming the parameter is unidentified. A practitioner can use this insight to adjust data collection or modeling assumptions accordingly. While straightforward, the method requires multiple training runs (ten models are trained in Figure 1), and we discuss how to reduce the computational cost in Section 4.

To summarize, we propose the use of profile likelihood analysis with confidence intervals to diagnose identifiability of two-tower models. Our work tackles the following three research questions:

RQ1: Does our profile likelihood diagnostic correctly detect identifiability from ranking swaps in controlled settings?

RQ2: How do feature overlap, ranking swaps, and sample size affect whether a two-tower model is practically identifiable?

RQ3: What identifiability problems emerge when applying our diagnostic to a real-world click dataset, such as Baidu-ULTR?

We first show that our method correctly diagnoses identifiability in models that require randomized document swaps to be identified (**RQ1**). We then demonstrate that our profile likelihood analysis distinguishes structural from practical unidentifiability, revealing that while feature overlap or randomized logging policies can resolve structural unidentifiability, practical unidentifiability persists without either sufficient feature overlap, randomization, or a large sample size (**RQ2**). Finally, we find that a simple two-tower model trained on a subset of the Baidu-ULTR dataset leaves several parameters unidentified, highlighting the motivation of this work and the practical need for a diagnostic method (**RQ3**).

Our contributions are:

- A profile likelihood-based diagnostic for identifiability in feature-based two-tower models trained on finite click data.
- An empirical distinction between structural and practical unidentifiability in two-tower models.
- A systematic simulation study of the effects of logging policies, feature overlap, and sample size on identifiability.
- An application to the Baidu-ULTR dataset demonstrating how to gain insight using our identifiability diagnostic.

The remainder of this paper is organized as follows. We formalize identifiability in two-tower models and introduce profile likelihood-based testing, followed by synthetic experiments analyzing the effects of logging policies, feature overlap, and sample size on model identifiability. We conclude by applying our method to the Baidu-ULTR dataset and discuss its practical implications.

2 Theoretical Background

In this section, we introduce two-tower models, review recent work on identifiability, and define the concept of profile likelihood.

2.1 Two-tower Models

Two-tower models predict the probability of a click $C \in \{0, 1\}$ for a search query $q \in Q$ and a document $d \in D$. We follow the simplified formulation of [14] and focus our notation on additive

two-tower models with position as a single bias feature, where $k \in K$ represents the rank of a document in the search results:

$$P(C = 1 \mid q, k, d) = \sigma(\theta_k + \gamma_{q,d}). \quad (1)$$

We use $\sigma(\cdot)$ to denote the sigmoid function. The model consists of two components: a *relevance tower* that outputs a logit $\gamma_{q,d}$ and a *bias tower* that outputs θ_k . The relevance tower predicts the utility of a document to a given query, independent of its display position. In practice, the relevance logit $\gamma_{q,d}$ may represent separate embedding parameters for each query-document pair or, more commonly, the output of a neural network $r(\cdot)$ applied to query-document features: $\gamma_{q,d} = r(x_{q,d})$, where $x_{q,d}$ captures ranking features such as BM25 scores or semantic embeddings [13, 40]. The bias tower computes a logit θ_k to capture position-dependent examination effects and can be extended to bias features beyond position [14]. During training, two-tower models jointly optimize both towers, commonly using the binary cross-entropy loss [7, 14, 40]:

$$\hat{\mathcal{L}}(\theta, \gamma \mid D) = -\frac{1}{N} \sum_{(q,d,k,c) \in \mathcal{D}} [c \log \sigma(\theta_k + \gamma_{q,d}) + (1-c) \log(1 - \sigma(\theta_k + \gamma_{q,d}))], \quad (2)$$

over a dataset of N observations: $\mathcal{D} = \{(q_i, d_i, k_i, c_i)\}_{i=1}^N$. During inference, however, query-document pairs are ranked based on their relevance logit $\gamma_{q,d}$. The bias tower effectively serves as a method to account for position bias during training, but is typically discarded during inference [11, 13, 40]. While our framework can be directly extended to more complex model architectures, for the remainder of this paper, we focus on a simplified two-tower model that considers only position bias.

2.2 Identifying Two-Tower Models

We define identifiability following Hager et al. [14], who base their definition on the work of Koller and Friedman [21, Definition 19.4]. A two-tower model is identifiable if no two distinct sets of parameters $(\theta_k, \gamma_{q,d})$, $(\theta'_k, \gamma'_{q,d})$ induce the same distribution over observable variables. This means that if:

$$P(C \mid q, d, k; \theta, \gamma) = P(C \mid q, d, k; \theta', \gamma') \quad (3)$$

it must follow that $\theta_k = \theta'_k$ for all positions and $\gamma_{q,d} = \gamma'_{q,d}$ for all query-document pairs. As a result, unidentifiability implies that we cannot uniquely recover model parameters from observed data as there are multiple (possibly infinite) parameter combinations that explain the observed data equally well [14, 21].

Chen et al. [5] investigate the identifiability of ULTR models following the examination hypothesis (i.e., models that assume that clicks can be factorized into bias and relevance, which two-tower models assume). Their work models relevance using a separate model parameter per query-document pair and shows that the relevance tower is identifiable when query-document pairs are observed across ranks, such that all positions form a connected graph [5]. Hager et al. [14] extend this framework to two-tower models using query-document features and show the fundamental problem of additive shifts in two-tower models, which we will restate in the following.

Suppose users actually behave according to the true bias and relevance parameters θ_k and $\gamma_{q,d}$ in Equation (1), and consider a

set of alternative parameters θ'_k and $\gamma'_{q,d}$. If both parameter combinations lead to the same observed click probability, then by the injectiveness of the sigmoid function $\sigma(\cdot)$, the following must hold:

$$\theta_k + \gamma_{q,d} = \theta'_k + \gamma'_{q,d}. \quad (4)$$

To show the inherent problem of identifying two-tower models, consider adding an arbitrary parameter shift $\Delta_k \in \mathbb{R}$ to all documents displayed at the same position [14]:

$$\theta_k = \theta'_k + \Delta_k, \quad \gamma_{q,d} = \gamma'_{q,d} - \Delta_k. \quad (5)$$

Note how we can arbitrarily choose Δ_k , while the predicted click probability stays the same. As a result, our model parameters can be adjusted freely without affecting the likelihood, and no unique optimal parameterization exists for the data, i.e., the model is not identifiable. To resolve this issue, consider observing a query-document pair at multiple positions. This constraint enforces equality of the corresponding Δ_k shifts across these positions. When such document swaps connect all positions, we have a single shared Δ_k [5, 14]. This last ambiguity is typically resolved through normalization, e.g., by setting the first bias parameter to $\theta_1 = 0$.¹ Finally, we obtain a single set of model parameters explaining our clicks.

Note an important property of the example above: once the bias parameters are identified, the relevance parameters are uniquely determined as well, as both vary with the same Δ_k . Thus, identifiability of one tower implies identifiability of the entire model [14].

The above example considers the identifiability of a model with separate relevance parameters for each query-document pair. However, recent work shows that two-tower models leveraging query-document features instead of unique identifiers can be identified from overlapping feature distributions across ranks (i.e., observing very similar documents across positions) [14]. In our experiments, we investigate diagnosing identifiability from randomized swaps and overlapping feature distributions in finite data settings. Next, we introduce profile likelihood as the foundation of our method.

2.3 Profile likelihood for identifiability

Profile likelihood has a decades-long history in assessing identifiability of maximum likelihood estimates (MLEs) and in constructing confidence intervals for statistical models [26, 33]. The profile likelihood of a parameter is obtained by fixing that parameter to a specific value and re-optimizing all remaining model parameters with respect to the original objective function [26, 29, 33]. This procedure yields a one-dimensional slice of the likelihood surface, reflecting how well the model can explain the observed data when the parameter of interest is constrained. Despite its long history, profile likelihood remains widely used to assess model identifiability in applied domains such as systems biology and bioinformatics, which inspired our approach [29].

Profile likelihood enables the construction of likelihood-based confidence intervals in finite samples. By Wilks' theorem, the likelihood ratio statistic asymptotically follows a chi-squared (χ^2) distribution [38]. Consequently, confidence intervals for a parameter can be derived by identifying values for which the profile likelihood does not exceed a threshold Δ_α , determined by the inverse cumulative distribution function (CDF) of a χ^2 distribution with one degree

¹Note that we suggest a different, constraint-based normalization in Section 4.2 as this simple normalization trick is not suitable for our setup.

of freedom at confidence level α [29]. Therefore, a parameter is considered identifiable if its profile likelihood exceeds this threshold in both increasing and decreasing parameter directions, yielding a finite confidence interval.

This framework allows to distinguish between structural and practical unidentifiability [29]. Structural unidentifiability arises from redundancies in the model structure itself, independent of data quantity or quality [23, 29, 36]. In this case, multiple parameter configurations yield identical model predictions, resulting in a flat profile likelihood that never exceeds the χ^2 threshold. Structural unidentifiability cannot be resolved by collecting more data, but requires modifying the model or the data collection process (e.g., introducing randomized swaps in two-tower models). In contrast, practical unidentifiability occurs when a parameter is structurally identifiable, i.e., its profile likelihood has a unique optimum, but the available data is insufficient to constrain it [37]. Practically unidentifiable parameters yield profile likelihoods that exceed the threshold in only one direction, leading to confidence intervals that are unbounded on one side [29]. Unlike structural unidentifiability, practical unidentifiability may be resolved by collecting additional or higher-quality data.

In practice, profile likelihood-based identifiability is assessed through a parameter-shifting procedure. Let P denote the set of optimized model parameters obtained by minimizing a loss function $\mathcal{L}$. For a parameter of interest $p \in P$, the profile likelihood is evaluated by fixing p to shifted values $p_{\text{shifted}} = p + s$, with $s \in \mathbb{R}$, while re-optimizing all remaining parameters. This yields a new parameter set $\hat{P}$ and a corresponding model loss. The resulting likelihood ratio test evaluates if the loss difference between the original and shifted model exceeds the threshold Δ_α :

$$\Delta_\alpha < [\log \mathcal{L}(P) - \log \mathcal{L}(\hat{P})]. \quad (6)$$

Although a χ^2 likelihood is commonly used during training [24, 29], Wilks' theorem allows significance testing to be applied to arbitrary likelihood functions [24, 38]. Evaluating this test across positive ($s > 0$) and negative ($s < 0$) shifts yields three behaviors corresponding to the definitions above [29, 37]: (i) if the threshold is never exceeded, the parameter is likely structurally unidentifiable; (ii) if the threshold is exceeded in only one direction, the parameter is practically unidentifiable; and (iii) if the threshold is exceeded in both directions, the parameter is identifiable.

3 Related Work

3.1 Identifiability in ULTR

The discussion of identifiability is a novel subject in the field of unbiased learning-to-rank (ULTR). Oosterhuis [27] highlighted that when logged rankings contain no swaps between certain ranks, clicks admit multiple bias-relevance combinations, and a click model trained on such data is not a consistent estimator of document relevance, i.e., even in infinite data, a structurally unidentified model is not guaranteed to converge to the true relevance parameters. Chen et al. [5] connect this issue to identifiability theory and show that document swaps need to form a connected graph of positions to identify models following the examination hypothesis. They also propose intervention methods that merge or edit bias features in the dataset to achieve identifiability. Research by

Hager et al. [14] has extended this analysis to additive two-tower models and shows that, under certain conditions, identifiability can be achieved not only from randomized swaps but also from overlapping document feature distributions across positions. These theoretical contributions are substantiated with simulations in synthetic settings with very large datasets [14]. However, how these conditions affect practical identifiability in smaller datasets, and how different degrees of feature overlap and randomized swaps interact, are underexplored. Additionally, previous work focuses on theoretical discussions of simplified models. Hager et al. [14] note, however, that different modeling choices (e.g., neural network depth or regularization) that are harder to analyze theoretically can also affect identifiability in practice. Our work provides a first step toward practically assessing the identifiability of two-tower models that practitioners use commonly but that are harder to analyze, e.g., a deep neural network using dropout regularization [13, 40].

3.2 Practical Identifiability

The framework by Raue et al. [29] remains effective and highly used in the field of bioinformatics for assessing the identifiability of individual parameters [25, 30, 37]. Recent work on profile-likelihood has investigated the use of multi-dimensional likelihood profiles to identify multiple, interdependent parameters [4]. Additionally, research by Joubert et al. [20] has developed a method to test identifiability without computing the full likelihood profile by using radial penalization [37]. Recent work in practical identifiability uses the definition of a model being identified if the confidence intervals of all estimated parameters are of finite size [37]. For any practically unidentified parameter, it is defined that it should be identified when given a sufficient amount of added data [37]. Model identifiability is considered to be verified experimentally if for all estimated parameters a finite confidence interval is found, i.e., all parameters are structurally and practically identifiable [37]. Our work provides a first application of profile likelihood to challenges of the unbiased learning-to-rank field and, thus, focuses on a simple method based on the general framework of Raue et al. [29].

4 Method

In this section, we present our method, which uses the profile likelihood to diagnose identifiability in two-tower models. Since the entire model is identified when one of the towers is identified (see Section 2.2), we focus on identifying the simpler tower, the position bias tower. Our method trains a two-tower model and then shifts individual bias parameters to observe if shifts in the resulting profile likelihood are statistically significant. In the following, we will first discuss how to determine if differences in profile likelihood are statistically significant. Then, we detail how to freeze and shift model parameters, before discussing how to distinguish between practical and structural identifiability. Finally, we detail how to move from identifying individual parameters to an entire model.

4.1 Assessing differences in profile likelihood

As covered in Section 2.3, we can assess if differences in profile likelihood are statistically significant by checking if the difference between two likelihoods exceeds a threshold derived from the χ^2

distribution.² Effectively, the result is a likelihood ratio test between the original model’s likelihood and the likelihood after shifting an individual parameter. Note, however, that this test requires unnormalized likelihoods. In contrast, loss functions of two-tower models are typically normalized by the number of samples [5, 14, 40]. To follow standard notation while preserving statistical validity, we use a normalized loss and scale the loss difference by the number of user sessions N in our dataset:

$$\Delta_\alpha < N |\mathcal{L}(P) - \mathcal{L}(\hat{P})|, \quad (7)$$

where $\mathcal{L}$ is the binary cross-entropy loss defined in Equation (2), $P = (\theta, \gamma)$ denotes the optimal parameters of the trained two-tower model, and $\hat{P}$ denotes the re-optimized parameters after shifting and freezing a single bias parameter. We refer to the expression $N |\mathcal{L}(P) - \mathcal{L}(\hat{P})|$ as the *unnormalized likelihood diff*.

4.2 Parameter Freezing and Shifting

Parameter Freezing. To compute the profile likelihood of a parameter, we take the original parameter value (e.g., a bias logit for a specific position) and shift it to a new value. The bias parameter is fixed (or frozen) at the new value while we re-optimize all remaining bias and relevance parameters. An unidentified model can compensate for this shifted bias parameter by adjusting the remaining parameters, whereas an identified model with only a single optimal parameterization cannot fully compensate for this shift (leading to a higher loss).

Note, however, that the identifiability theory for two-tower models concerns the output logits of the relevance and bias towers (Section 2.2, not their internal parameters (i.e., not the weights and biases of a neural network). Thus, our freezing operation must ensure that, e.g., the bias tower returns a fixed logit for a specific position, while allowing the rest of the model to adjust freely.

Therefore, we implement shifting and freezing a bias logit by simply bypassing the bias tower for the frozen position. During each forward pass, we output the shifted, fixed value for the frozen position and infer the bias tower for all other positions as usual. This ensures that the model can optimize all other parameters while the shifted value remains fixed throughout training.

Normalization. Shifting individual parameters introduces a subtle challenge for conventional two-tower implementations. As detailed in Section 2.2, two-tower models are identifiable only up to an additive constant, an ambiguity typically resolved by normalization (e.g., fixing the first bias logit to $\theta_1 = 0$ [14]). However, this approach is incompatible with parameter shifting: we cannot shift a parameter constrained to zero, and any shift in other parameters can be absorbed by adjusting the relevance tower by the same amount, leaving click probabilities unchanged.

To address this, we propose to use mean normalization during optimization [14], constraining the mean of all bias logits to zero rather than fixing a single parameter. This preserves a unique parameterization while allowing each parameter to be shifted: a positive shift represents an increase relative to the mean, and a negative shift, a decrease. We implement this by adding a penalty term

$\left(\frac{1}{K} \sum_{k=1}^K \theta_k\right)^2$ to the training loss in Equation (2), which encourages the bias logits to remain centered at zero throughout optimization.

Shifting Factor. Recall that the shift factor s determines how far each bias parameter is moved from its optimal value. While profile likelihood is defined over the entire parameter space, evaluating it requires re-optimizing the model at each shift value, so we must balance comprehensive evaluation with computational cost.

Fortunately, for two-tower models, we can bound the useful range of shifts. Since click probabilities lie in $[0, 1]$, extreme parameter values yield diminishing returns: the sigmoid activation saturates, and further shifts have a negligible effect on predictions. For the two-tower formulation in Section 2.1, a shift of $s = 3$ already pushes the sigmoid output beyond 95% saturation, making larger shifts unlikely to reveal additional changes in the likelihood.

In practice, evaluating identifiability requires only a few shifts. For a well-behaved likelihood with a clear optimum, a single large positive and negative shift (e.g., $s \in \{-3, +3\}$) often suffices. If these shifts produce significant loss increases in both directions, the parameter is identified. The total number of training runs is the number of bias parameters (e.g., positions) times the number of shift values evaluated per parameter. For example, with 10 positions and two shifts per parameter, this yields 20 training runs. Please note that other model architectures (e.g., models without a sigmoid activation) may require different parameter shifting tactics [29].

4.3 Diagnosing identifiability types

Our proposed diagnostic can help to differentiate the type of unidentified practitioners may encounter. We follow the approach from previous work [29] (as described in Section 2.3) and classify a parameter as structurally unidentified if none of the studied shifting factors exceeds the identifiability threshold (e.g., at a confidence level of $\alpha = 0.05$). We classify a parameter as practically unidentified if the identifiability threshold is exceeded for one of the shifting directions (e.g., positive or negative), but not in the other shift direction. As indicated in previous work, structural unidentifiedness is caused by redundancies in the model parameterization or insufficient variability in the data collection process, such as a lack of document swaps across positions [29, 37]. Practical unidentifiedness arises from insufficient data, and can typically be resolved by collecting more samples [29]. A parameter is fully identified when both shifting directions exceed the identifiability threshold.

4.4 From parameter to model identifiability

As discussed in Section 2.2, identifiability of the bias parameters implies identifiability of the entire two-tower model, since both towers share the same additive ambiguity. The profile likelihood method tests identifiability for each bias parameter separately, yielding a separate hypothesis test for each parameter. However, when testing multiple parameters simultaneously, we must account for the multiple comparisons problem: conducting n_k independent tests at significance level α increases the probability of falsely concluding that at least one parameter is identifiable.

To control for this, we apply the Bonferroni correction [31]. Rather than testing each parameter at level α , we use the adjusted

²From the the inverse CDF of the χ^2 distribution with a single degree of freedom.

significance level $\alpha_{\text{new}} = \frac{\alpha}{n_k}$, where n_k is the number of bias parameters (e.g., positions) and α is the desired significance level. We then compute the corresponding χ^2 threshold $\Delta_{\alpha_{\text{new}}}$ using this stricter significance level. If *all* bias parameters exceed this threshold in both shift directions, we conclude that the full model is identifiable at significance level α . This approach is consistent with prior work, which considers a model identifiable when all individual parameters are identified [24, 37, 38].

5 Experiments

In the following, we validate our profile likelihood-based diagnostic through a series of experiments in controlled synthetic settings and in real-world data. We address three research questions that systematically evaluate different aspects of identifiability in two-tower models. First, we verify that our method can detect the effects of ranking swaps on identifiability in a synthetic setting (RQ1). Second, we investigate how the degree of feature overlap and sample size affect both structural and practical identifiability, and investigate our method’s ability to distinguish between these two types of unidentifiability (RQ2). After validating our method in simulations, we apply our diagnostic to the Baidu-ULTR dataset to test whether it can uncover identifiability issues in real-world data (RQ3).

5.1 RQ1: Identifiability from ranking swaps

To verify our proposed method for measuring identifiability, we first experiment in a simulation setup in which query-document pairs are represented by unique one-hot encoded feature vectors (i.e., every query-document pair is represented by a unique $y_{q,d}$). This design ensures that documents cannot be connected via shared features, meaning identifiability can arise only from observing the same documents at different positions. This controlled setting serves as a sanity check, verifying that our method can reliably distinguish between identifiable and unidentifiable scenarios: From work in [5] and [14], we expect to detect identifiability when documents are randomly swapped across positions, but not when rankers are deterministically showing the same documents always in the same positions.

5.1.1 Experimental Setup.

Dataset. We construct a synthetic dataset with 100 one-hot-encoded features, representing 10 queries, each with 10 documents. Each query-document pair is assigned a discrete relevance label sampled uniformly at random from 1 to 5. We train a relevance tower on the complete ground-truth relevance annotations of the training set with a mean-squared error loss, following Hager et al. [14]. We simulate a large dataset of 1M clicks for training and 500k for validation and testing, respectively. Meaning that for each query, our model effectively encounters all click permutations.

Model. Following previous work [14], we use a simple two-tower model with a 2-layer feed-forward neural network as the relevance tower (2 layers, 32 neurons, ELU activations). And a single parameter per rank, as a position bias logit for the bias tower. All models were optimized using AdamW [22] with a learning rate of 0.003 and a weight decay of 0.01 for up to 50 epochs, with early stopping after three epochs without improvement of the validation loss.

Logging policy. For this experiment, we require a logging policy that can simulate a setup without any document swaps (i.e., a deterministic policy) and can gradually introduce ranking swaps. For this, we reuse the epsilon-greedy policy proposed in [14]. With probability T , we sample a uniform ranking of documents. Otherwise, we rank deterministically based on our ground-truth relevance labels. We refer to probability T as the policy’s *temperature*. A policy with $T = 0$ never swaps a document across positions, whereas a policy with $T = 1$ uniformly swaps documents across all positions.

5.1.2 Results. We investigate the concept of identifiability in a highly synthetic setting in which all query-document pairs are independent. We qualitatively visualize the profile likelihood for the first two position bias parameters and explore shift factors $s \in \{-3, -2, \dots, 3\}$ across four policy temperatures $T \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$.

The results of this experiment are shown in Figure 2. As expected, when training two-tower models on deterministic rankings ($T = 0$) without any swaps, all parameter shifts result in the same log loss when compared to the original model parameters. The flat profile likelihood correctly indicates that the parameter is structurally not identifiable, as the model can compensate for the shifted parameter by adjusting other parameters, achieving the same optimal loss.

When considering models trained on data collected with non-zero temperature settings, there is a clear increase in log loss for larger shifts s . We also observe that a higher policy temperature leads to a stronger increase in log loss, indicating that models trained on more randomized data are easier to identify.

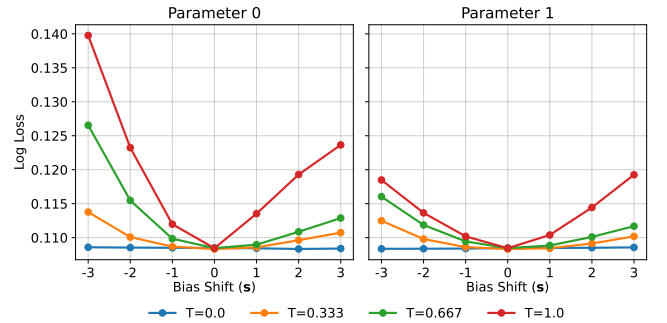


Figure 2: Test loss when shifting the first two bias parameters by $s \in \{-3, -2, \dots, 3\}$ across four logging policy temperatures.

5.2 RQ2: Identifiability from features

Having established that our method can detect identifiability effects from ranking swaps (RQ1), we now investigate how identifiability manifests in simple feature-based models, i.e., when each query-document pair is represented through a vector $x_{q,d}$ used by the relevance tower: $y_{q,d} = r(x_{q,d})$.

Recent work shows that two-tower models can be identified through overlapping query-document features across positions [14] even without randomized swaps. However, these theoretical guarantees assume infinite data. In practice, we expect identifiability to depend on both the *degree of feature overlap* across positions and the *amount of available data*. When each rank shows documents from completely distinct feature ranges (no overlap), we expect two-tower models not to be identified from features alone. When feature

distributions overlap across ranks, documents with similar features appear at multiple positions, potentially enabling identification. In **RQ2**, we will investigate the interaction between feature overlap, randomized swaps, and training dataset size on identifiability.

5.2.1 Experimental Setup. We use the same simulation, model, and logging policy setup as described in Section 5.1. Only the setup of the relevance tower deviates, as we use a deeper feed-forward neural network (5 layers, 128 neurons, ELU activations).

Dataset. To better visualize our experiments, we focus on a single continuous query-document feature, which we sample uniformly from predefined subranges per rank to control feature overlap. The overlap parameter controls the spacing between these ranges: negative values create separation (no overlap), positive values create intersection between adjacent ranks, and large values create full overlap where all ranks sample from the same distribution. To isolate our setup from expert labels, we generate synthetic relevance labels by passing the feature through a randomly initialized 3-layer neural network (32 neurons per layer, tanh activations) to simulate a non-linear relevance function that our relevance tower needs to recover (similar to [14]).

To simulate different dataset sizes, we sample 100 user sessions with clicks per query for training and vary the number of queries (and thus, overall query-document pairs). This enables us to distinguish structural unidentifiability (due to insufficient overlap) from practical unidentifiability (due to insufficient data points). We simulate 10 user sessions per query for validation and testing, respectively. Lastly, we plot the chi-squared thresholds for three significance levels $\alpha \in \{0.001, 0.01, 0.05\}$.

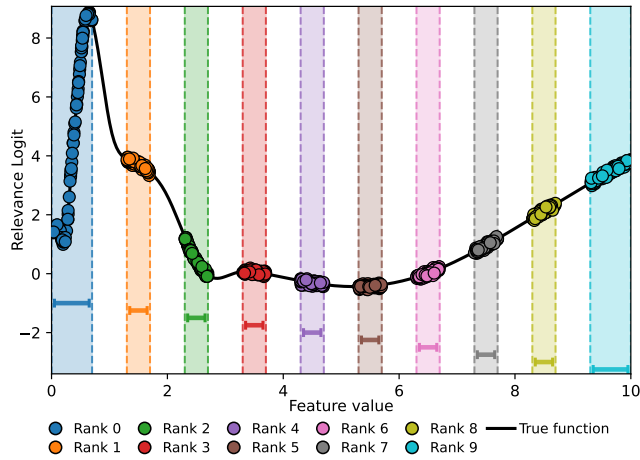


Figure 3: Sample value distribution for a score function where there is a separation between different ranks, 100 query samples are drawn with a separation of -0.3 between ranks.

5.2.2 Structural Unidentifiability. First, we investigate identifiability when there is no feature overlap between ranks. In this setting, each rank occupies a distinct region of the feature space. Figure 3 illustrates the simulated dataset, where each point is a query-document pair with a given value and relevance score. Note how, regardless of the number of query-document pairs we sample,

no area of the feature space is shared across two ranks. From recent work, we expect a two-tower model to be structurally unidentified in this setting, as models can learn arbitrary shifts for documents at each rank [14].

Figure 4 displays the profile likelihood for the first bias parameter. Under the deterministic logging policy with temperature $T = 0$, we observe an erratic (but overall flat) profile likelihood independent of the number of queries. Notably, the model trained on the largest dataset with 500 queries is not closer to being identified than a model trained on 10 queries (we do observe some noise for the model trained on 100 queries). This indicates that our parameter is structurally unidentified due to a lack of feature overlap across ranks, and training on more data will not resolve this problem. Note also the lack of a unique optimum in the profile likelihood across all datasets as another indicator of structural identifiability problems.

For the models trained on data collected by a uniform random policy with $T = 1$, we expect from theory that models are identified from randomized swaps (Section 2.2). However, while we observe hints of a single optimum in the profile likelihood, the models trained on 10 and 25 queries fail to exceed the likelihood threshold. In other words, while theory implies structural identification, we observe that the model is practically not identified from finite data. Here, it should be noted that not finding identifiability in settings with few samples is also a feature of applying our statistical threshold, given that the unnormalized likelihood diff is scaled by the number of samples. However, for 100 and 500 queries, we observe that the loss differences exceed the significance thresholds in both shifting directions, indicating that the models are both structurally and practically identified in Figure 3.

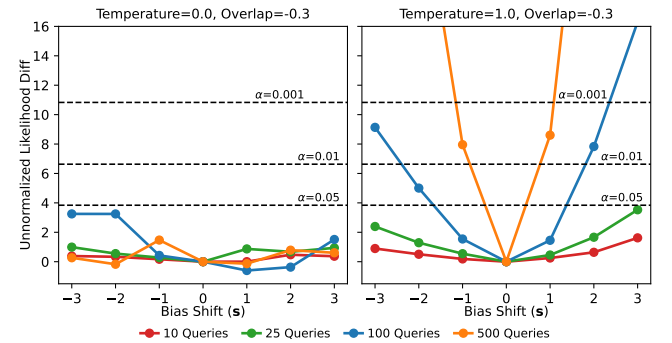


Figure 4: Identifiability threshold comparison for the first bias parameter under the profile likelihood for 10, 25, 100, and 500 queries when dataset features are drawn with a full separation (0.3) between different ranks. We use a deterministic ($T = 0.0$) and random logging policy ($T = 1.0$).

5.2.3 Practical Unidentifiability. Next, we examine a scenario with moderate feature overlap between ranks. Theoretically, when ranks are connected through feature overlap, the model should be identifiable even without randomized swaps [14]. Figure 5 shows the feature distribution for this setting. While feature ranges of adjacent ranks overlap (e.g., ranks 0 and 1 can both share documents with features around 0.6), the actual observed samples may not be close enough in feature space to constrain the model. For instance,

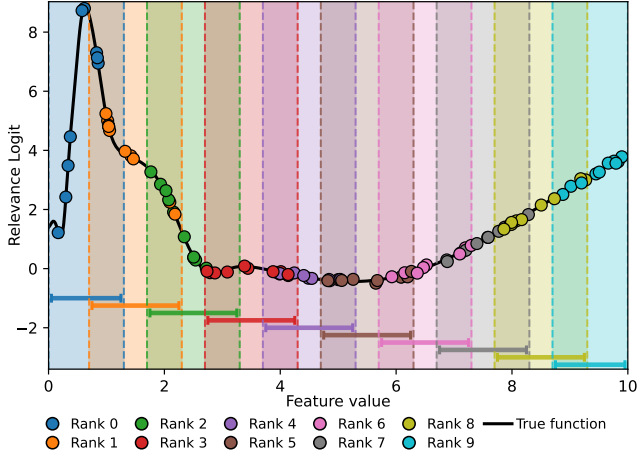


Figure 5: Sample value distribution for a score function where there is a separation between different ranks, 10 query samples are drawn with an overlap of 0.3 between ranks.

rank 0 samples cluster near 0.4 while rank 1 samples cluster near 0.7, creating only weak connections despite overlapping ranges.

The results in Figure 6 confirm this concern. Under the deterministic logging policy ($T = 0$), we observe increasing loss when evaluating identifiability on the dataset from 10 to 100 queries, but the differences do not reach the significance thresholds. The model is structurally identifiable (there exists a unique optimum), but the available data is insufficient to reach a significant difference, i.e., the model is not practically identifiable. With 500 queries, we are able to reach the identifiability threshold under the largest shift. Under the random logging policy ($T = 1$), the model becomes identified after training on 100 queries, highlighting the importance of sample size and swap interventions for identifiability even if document features overlap in theory.

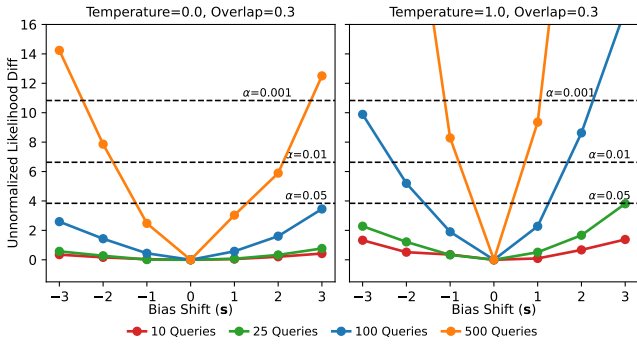


Figure 6: Identifiability threshold comparison for the first bias feature under the profile likelihood for 10, 25, 100, and 500 queries when dataset features are drawn with a limited overlap (0.3) between different ranks. We show results for a deterministic ($T = 0.0$) and random logging policy ($T = 1.0$).

5.2.4 Full feature overlap. Finally, Figure 7 examines identifiability under full overlap where all ranks sample from the same distribution. Under the deterministic logging policy, a model trained on

up to 100 queries is unidentified, mirroring our results from the limited feature overlap setting. With 500 queries we achieve full identification. Notable when comparing to the setting with limited overlap is that identifiability is reached with a smaller shift (e.g., for $\alpha = 0.01$ at 500 queries). Finally, models trained on the random logging policy are already identified at 100 queries, emphasizing that swaps improve identification even with full overlap.

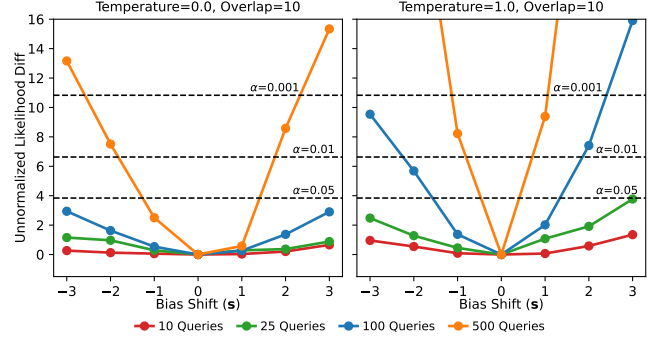


Figure 7: Identifiability threshold comparison for the first bias feature under profile likelihood for 10, 25, 100, and 500 queries when dataset features are drawn uniformly from the same distribution. We show results for a deterministic ($T = 0.0$) and random logging policy ($T = 1.0$).

Our results demonstrate that our profile likelihood method effectively diagnoses identifiability across three regimes: structural unidentifiability (no overlap), practical unidentifiability (moderate overlap or insufficient samples), and full identification (sufficient overlap and sample size). Adding nuance to recent work [14], our experiments highlight the practical benefit of randomized swap interventions for identifying two-tower models even when generalizing across overlapping query-document features.

5.3 RQ3: Identifiability in Baidu-ULTR

As our previous experiments were conducted in simplified simulations, our final experiment (RQ3) applies our method to investigate identifiability on a large real-world dataset, Baidu-ULTR [13, 43].

5.3.1 Experimental setup. We use a preprocessed subset of the Baidu-ULTR dataset published in [13]. We use one partition of data, including $\approx 590k$ user sessions, $\approx 4.8M$ query-document pairs, with 13 learning-to-rank features (e.g., BM25, TF-IDF, etc.) and use position as our only bias feature.

We train a two-tower model according to the best hyperparameters in [13], using a relevance tower with 5 layers, 128 neurons, and ELU activations, and optimized with AdamW [22].

5.3.2 Results. We apply our framework to position bias in Baidu-ULTR evaluating four shifts per parameter. We display our results in a practical CLI tool as shown in Figure 8. The CLI displays all evaluated parameters, upward (Pos Shift) and downward (Neg Shift) shifts, the reached significance level, and resulting conclusions. We also report the largest change in loss per parameter and the number of training samples (i.e., user sessions with clicks). A low sample count can indicate that a position is under-represented in

the dataset, e.g., because users might not scroll far enough down a list of search results.

Figure 8 shows interesting identifiability results: Higher positions are generally identified, and parameters beyond position 14 are unidentified. Interestingly, several parameters are found to be practically unidentified. This example emphasizes the usefulness of diagnosing identifiability in practice, as some parameters (e.g., at position five) are practically unidentified despite having a similar number of samples to adjacent positions. This could indicate a problem in the query-document features distribution, e.g., due to a certain item being always displayed in that position.³

Param	p-val	Neg Shift	Pos Shift	Max Diff Loss	Samples	Conclusion
0	0.0001	-1.5	3.0	0.001853	2501890	identified
1	0.0001	-1.5	3.0	0.001022	2430155	identified
2	0.0001	-1.5	3.0	0.000301	2575640	identified
3	0.0001	-3.0	3.0	0.000391	2570180	identified
4	0.0001	-3.0	3.0	0.003672	2595790	identified
5	0.0001	-1.5	None	0.002680	2624070	practically unidentified
6	0.0001	-3.0	1.5	0.000660	2745480	identified
7	0.0001	-1.5	1.5	0.001908	2606895	identified
8	0.0001	-1.5	None	0.000839	2782795	practically unidentified
9	0.0001	-1.5	None	0.000458	342855	practically unidentified
10	None	None	None	0.000532	121980	unidentified
11	0.0001	-3.0	None	0.003546	112575	practically unidentified
12	0.0001	-1.5	None	0.000894	81240	practically unidentified
13	0.05	-1.5	3.0	0.000884	57825	identified
14	None	None	None	0.000216	35200	unidentified
15	None	None	None	0.000366	19535	unidentified
16	None	None	None	0.000166	10595	unidentified
17	None	None	None	0.000374	6725	unidentified
18	None	None	None	0.000555	3095	unidentified
19	None	None	None	0.000300	135	unidentified

Conclusion: The model-dataset combination is not identified for this dataset.

Figure 8: Output of our identifiability CLI tool applied to 20 position bias parameters on Baidu-ULTR. Each row reports the significance level reached, the largest significant shift in each direction, and the resulting identifiability conclusion.

We plot the same results in Figure 9. At the top, we show identifiability results per parameter, plotting a point for the largest loss difference for the positive and negative parameter shift. We color each dot by the corresponding identifiability conclusion. Note that when a dot is not visible, the loss difference is negative (which may occur if the original model converges to a local optimum).

The bottom panel of Figure 9 reveals the likely underlying cause: sample counts remain stable for positions 0-8 (reflecting the average of eight documents per result page in the Baidu-ULTR subset [13]) but drop sharply thereafter. This severe imbalance likely prevents positions 10-19 from being identified due to insufficient data and may create a disconnect between early and late positions, undermining the identifiability of the entire model.

6 Practical Implications and Limitations

Our work has several practical implications: first, we show that satisfying distributional constraints of a dataset, such as overlap between ranks or features as described in previous literature [5, 14], does not always guarantee identifiability of a two-tower model in finite datasets. The method presented in this work expands the toolbox of practitioners with a computationally tractable approach that can detect whether a given model is structurally and practically identifiable on a given dataset. As our work is the first to adopt

³Previously, not removing query suggestion boxes has caused trouble on Baidu-ULTR, which might be a potential explanation for our observation [13].

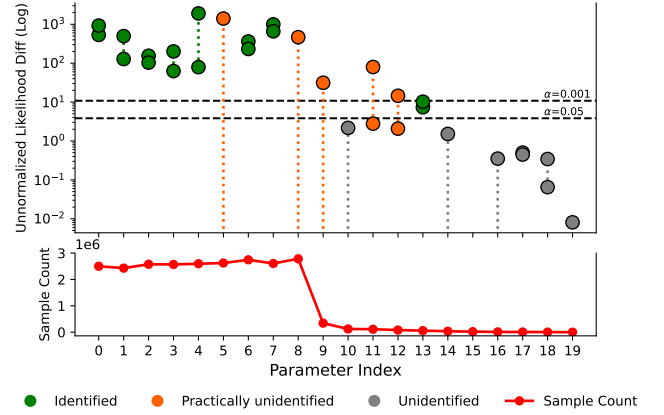


Figure 9: Identifiability results for 20 position bias parameters on Baidu-ULTR, showing the unnormalized likelihood difference across positive and negative shifts. Points are colored by identifiability conclusion. Sample counts per position are shown below, revealing a sharp drop beyond position 8 that coincides with the onset of unidentified parameters.

profile likelihood for diagnosing identifiability in unbiased learning-to-rank, it comes with several limitations.

While our methods can diagnose identifiability, we do not propose a method in this work to mitigate parameter unidentifiability. Future work may explore changes to two-tower models (such as regularization or shallower towers) or changing data properties or collection processes to resolve identifiability problems. One concrete direction is to investigate the trade-off between logging policy randomization and the amount of data needed to reach identifiability, as our experiments suggest that more aggressive randomization reduces the sample size required for practical identification. Future work could formalize this relationship and use it to guide data collection decisions in practice.

In addition, the feature overlap experiments in this work are limited to single continuous or one-hot encoded features. Future work could extend these experiments to high-dimensional feature representations and more realistic data distributions, testing how different levels of overlap interact with randomization in practice. A further open question is how the profile likelihood thresholds for individual parameters relate to the graph connectivity condition for identifiability described in previous work [5, 14]. Chen et al. [5] require all positions to form a connected graph through shared documents, yet our diagnostic tests each bias parameter in isolation. Future work could investigate how local identifiability of individual parameters aggregates to the global connectivity condition, and what minimum level of feature overlap between adjacent positions is sufficient to connect the full position graph in practice.

Furthermore, this work focused on simple additive two-tower models with a position bias tower. Future work should explore alternative model architectures and practical diagnostics for identifiability in models that use bias features beyond position [6, 40]. While our method can be directly adopted to multiple bias features, our current exhaustive parameter evaluation may quickly become prohibitively expensive, requiring more intelligent exploration strategies. Additionally, for evaluating multiple bias parameters, consideration must be given to feature overlap across bias

features. While each bias feature might be identified on its own, it may not be enough to diagnose limited overlap in the joint distribution of bias features [14].

Finally, future work should consider how to adapt profile likelihood-based methods to evaluate unidentifiability in more complex two-tower model setups that may require joint parameter shifting.

7 Conclusion

In this work, we introduce a method to diagnose the identifiability of additive two-tower models for unbiased learning-to-rank based on profile likelihood. By inspecting how model fit changes as individual parameters are shifted away from their optima, we provide a simple, computationally tractable diagnostic metric that allows testing the identifiability of a specific model architecture on a given dataset.

We first validated our method in controlled simulations, showing that identifiability in two-tower models manifests as a curvature in the profile likelihood of individual parameters and that finite-data effects, such as limited randomization, limited overlap between feature distributions, or limited amount of samples, can lead to practical identifiability problems, even when structural conditions for identifiability are satisfied in theory. By detecting identifiability issues on the real-world dataset Baidu-ULTR, we emphasize that identifiability is not solely a theoretical problem, but is strongly influenced by modelling assumptions, dataset collection and size.

Our work bridges the gap between theoretical identifiability discussions and practical diagnosis, offering a concrete pathway to assessing identifiability of additive two-tower models in ULTR.

Acknowledgments

Philipp Hager was funded by the Mercury Machine Learning Lab created by TU Delft, the University of Amsterdam, and Booking.com. Stan Fris was financially supported by Qualcomm and the European Laboratory for Learning and Intelligent Systems (ELLIS) Unit Amsterdam, and received a SIGIR Student Travel Grant. All content represents the opinion of the authors, which their employers or sponsors do not necessarily endorse.

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